A Higher-Regularity Cochain Extension by Contracting Homotopy

Automated Math Discovery Candidate Proof

August 9, 2026

Status. Candidate full solution to Open question 1 of Nilsson–Pitassi, likely valid pending expert review. The result answers the literal fixed-domain boundedness question. It does not claim a sharp bound uniform over varying families of Lipschitz domains.

Source Question

Nilsson and Pitassi ask in Section 6, Open question 1, pages 19–20 of arXiv:2604.04927v2 [1], whether their cochain extension has a strict higher-regularity analogue. For $m \in \mathbb{N}_0$, put

$$\mathcal{X}^k_m(\Omega) := H^{(m,m)}\Lambda^k(\Omega) = \{\omega \in H^m\Lambda^k(\Omega) : d\omega \in H^m\Lambda^{k+1}(\Omega)\}$$

with its graph norm, and

$$\mathcal{C}^k_m(\Omega) := \mathcal{X}^k_m(\Omega) \cap \mathfrak{H}^k(\Omega)^\perp,$$

where $\mathfrak{H}^k(\Omega)$ is the absolute L^2 harmonic space used in the source paper. The question asks for bounded operators

$$E^k_m : \mathcal{C}^k_m(\Omega) \longrightarrow \mathcal{X}^k_m(K)$$

which extend the datum and commute strictly with d. It also asks whether, when $\Omega \Subset K$, the extensions can have vanishing outer traces and therefore extend by zero to the whole space.

$$
\begin{aligned}
C_U &\lesssim C_P(\mathrm{Conv}(\Omega_+))\big(2 + C_{P,0}(\mathrm{Conv}(\Omega_+))C_{\mathrm{HLZ}} + C_{\mathrm{tr}}(A)\, C_{\mathrm{tr}}(\Omega)\big) \\
&\lesssim \mathrm{diam}(\mathrm{Conv}(\Omega_+))^2(C_{\mathrm{HLZ}} + C_{\mathrm{tr}}(A)\, C_{\mathrm{tr}}(\Omega)) =: \mathrm{diam}(\mathrm{Conv}(\Omega_+))^2/C.
\end{aligned}
$$

6 Further discussion and open questions

We conclude by discussing several open problems and possible future directions suggested by the construction and applications developed in this paper.

Our graded extension family is only bounded in $H\Lambda^k(K)$. Even when Ω and K are smooth, elliptic regularity applied to the first-order potential problem, equivalently to the gauge formulation in Section 3.4, yields improved regularity only on each side of the interface $\partial\Omega$. Recovering higher regularity across $\partial\Omega$ would require additional compatibility conditions. This leads naturally to the question of whether Theorem 1 admits a strict higher-regularity analogue.

Let $m \in \mathbb{N}_0$. For a bounded Lipschitz domain $\Omega \subset \mathbb{R}^n$ and form degree k, define $H^m\Lambda^k(\Omega)$ as the Sobolev spaces of L^2-integrable differential forms with coefficients in $H^m(\Omega)$, with convention $H^0(\Omega) = L^2(\Omega)$. Then define

$$
H^{(m,m)}\Lambda^k(\Omega) := \big\{\omega \in H^m\Lambda^k(\Omega) : d\omega \in H^m\Lambda^{k+1}(\Omega)\big\}, \tag{16}
$$

equipped with the graph norm

$$
\|\omega\|^2_{H^{(m,m)}\Lambda^k(\Omega)} := \|\omega\|^2_{H^m\Lambda^k(\Omega)} + \|d\omega\|^2_{H^m\Lambda^{k+1}(\Omega)}.
$$

See [19, Section 2] for more details.

Open question 1. For simplicity, let us first focus on the variant in Theorem 1 (b). Can one construct, for each form degree k, a bounded operator

$$
E^k_m : H^{(m,m)}\Lambda^k(\Omega) \cap \mathfrak{H}^k(\Omega)^\perp \to H^{(m,m)}\Lambda^k(K)
$$

such that, for every $\omega \in H^{(m,m)}\Lambda^k(\Omega) \cap \mathfrak{H}^k(\Omega)^\perp$,

(i) $E^k_m\omega|_\Omega = \omega$,

(ii) $\|E^k_m\omega\|_{H^{(m,m)}\Lambda^k(K)} \le C\, \|\omega\|_{H^{(m,m)}\Lambda^k(\Omega)}$,

(iii) $d\, E^k_m\omega = E^{k+1}_m(d\omega)$ in K?

If, in addition, Ω is strictly contained in K, can one impose vanishing tangential traces for all directions on ∂K so that the corresponding extension by zero defines an operator into $H^{(m,m)}\Lambda^k(\mathbb{R}^n)$?

The lower bound in Corollary 18 is not asymptotically optimal in the power of the diameter. This loss comes from the fact that the argument uses the Poincaré constant twice. At present, it is not clear how to remove the second occurrence, which appears in the proof of Theorem 15. An alternative route would be to combine our construction with a whole-space lifting such as [19, Lemma 5.1], but the low-frequency part of the estimate seems in general to introduce a dependence on $\mathrm{diam}(K)^{n/2}$. Another

Proof Intuition

The construction separates the problem into an internal homological step on Ω and an arbitrary degreewise Sobolev extension. Maximal regularity for the de Rham complex implies that the harmonic-orthogonal graph-Sobolev complex $\mathcal{C}^\bullet_m(\Omega)$ is exact. Since it is an exact Hilbert complex with closed differentials, it has a bounded contracting homotopy h, meaning

$$
dh + hd = I.
$$

Let S^k be any bounded extension in degree k; it need not commute with d. The two-term formula

$$E^k_m = dS^{k-1}h^k + S^k h^{k+1}d$$

repairs it. Restriction to Ω gives $dh + hd = I$, while applying d to the formula makes the cochain identity automatic because $d^2 = 0$.

The only delicate point is exactness at higher regularity. An exact H^m form has an H^{m+1} potential by Costabel–McIntosh. That potential need not initially be orthogonal to the L^2 harmonic fields, but their smooth cohomology representatives allow a finite-dimensional closed correction which enforces the orthogonality without changing the derivative.

Main Result

Theorem 1. *Let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz open set and let $K \subset \mathbb{R}^n$ be a bounded open set containing Ω. For every $m \in \mathbb{N}_0$ and $0 \le k \le n$, there is a bounded linear operator*

$$E^k_m : \mathcal{X}^k_m(\Omega) \cap \mathfrak{H}^k(\Omega)^\perp \longrightarrow \mathcal{X}^k_m(K)$$

such that

$$(E^k_m\omega)|_\Omega = \omega,$$
$$dE^k_m\omega = E^{k+1}_m(d\omega).$$

Here the spaces in degrees outside $\{0, \dots, n\}$ and the corresponding operators are understood to be zero.

If $\Omega \Subset K$, the operators may be chosen so that every $E^k_m\omega$ is supported in a fixed compact subset of K. Consequently all outer traces which occur in the graph-Sobolev setting vanish, and extension by zero defines a bounded operator into $H^{(m,m)}\Lambda^k(\mathbb{R}^n)$.

Remark 1. *The theorem is slightly stronger topologically than the question: the construction does not use any condition on $K \setminus \overline{\Omega}$. Its norm is finite for each fixed domain pair. No quantitative uniformity over varying domain families is asserted.*

The Harmonic-Orthogonal Sobolev Complex

We use two standard facts. First, the L^2 Hodge decomposition in the notation of the source paper is

$$L^2\Lambda^k(\Omega) = \mathfrak{H}^k(\Omega) \oplus d(H\Lambda^{k-1}(\Omega)) \oplus \ker(d|_\Omega)^\perp. \tag{1}$$

Second, Costabel and McIntosh prove maximal regularity and regularity of cohomology on bounded Lipschitz domains [2, Corollary 4.7 and Theorem 4.9]. In particular:

1. if $z \in H^m\Lambda^k(\Omega)$ is exact with an L^2 potential, then there is $v \in H^{m+1}\Lambda^{k-1}(\Omega)$ with $dv = z$;
2. there is a finite-dimensional space $\mathcal{G}^k \subset C^\infty(\overline{\Omega}, \Lambda^k)$ of closed forms which represents de Rham cohomology at every Sobolev order.

Lemma 1. *For every $m \in \mathbb{N}_0$, the spaces $\mathcal{C}^k_m(\Omega)$ form an exact Hilbert complex under d.*

Proof. The space $\mathcal{C}^k_m(\Omega)$ is closed in the Hilbert graph space $\mathcal{X}^k_m(\Omega)$. If $q \in \mathfrak{H}^{k+1}(\Omega)$ and $\omega \in \mathcal{C}^k_m(\Omega)$, the Green identity used in the source paper gives

$$\langle d\omega, q\rangle_{L^2} = \langle \mathrm{t}\,\omega, \mathrm{n}\,q\rangle_{\partial\Omega} - \langle \omega, \delta q\rangle_{L^2} = 0.$$

Thus $d\omega \perp \mathfrak{H}^{k+1}(\Omega)$, and $\mathcal{C}^\bullet_m$ is a complex.

Let $z \in \mathcal{C}^k_m(\Omega)$ satisfy $dz = 0$. By (1), the two conditions $dz = 0$ and $z \perp \mathfrak{H}^k(\Omega)$ imply that $z = du_0$ for some $u_0 \in H\Lambda^{k-1}(\Omega)$. Costabel–McIntosh maximal regularity, applied with Sobolev indices $s = m$ and $t = 0$, yields $v \in H^{m+1}\Lambda^{k-1}(\Omega)$ with $dv = z$. In particular $v \in \mathcal{X}^{k-1}_m(\Omega)$.

It remains to impose $v \perp \mathfrak{H}^{k-1}(\Omega)$. Let $P_{\mathfrak{H}}$ be the L^2 orthogonal projection onto $\mathfrak{H}^{k-1}(\Omega)$. The restriction

$$P_{\mathfrak{H}}|_{\mathcal{G}^{k-1}} : \mathcal{G}^{k-1} \longrightarrow \mathfrak{H}^{k-1}(\Omega)$$

is an isomorphism. Indeed, both spaces have dimension equal to the relevant Betti number. If $g \in \mathcal{G}^{k-1}$ has $P_{\mathfrak{H}} g = 0$, then g is closed and harmonic-orthogonal, so (1) makes it exact. Maximal regularity gives a Sobolev potential, and the direct-sum property of $\mathcal{G}^{k-1}$ in [2, Theorem 4.9] forces $g = 0$.

Set

$$g = (P_{\mathfrak{H}}|_{\mathcal{G}^{k-1}})^{-1} P_{\mathfrak{H}} v, \qquad u = v - g.$$

Then g is smooth and closed, hence $u \in \mathcal{C}^{k-1}_m(\Omega)$, and $du = dv = z$. This proves exactness. At degree zero, a closed function is locally constant and therefore belongs to $\mathfrak{H}^0(\Omega)$, so the same statement says that the harmonic-orthogonal cycle space is zero. □

A Bounded Contracting Homotopy

Lemma 2. *There are bounded linear maps*

$$h^k : \mathcal{C}^k_m(\Omega) \longrightarrow \mathcal{C}^{k-1}_m(\Omega)$$

such that

$$dh^k + h^{k+1}d = I \quad \text{on } \mathcal{C}^k_m(\Omega). \tag{2}$$

Proof. Let

$$\mathcal{Z}^k_m = \ker(d : \mathcal{C}^k_m \to \mathcal{C}^{k+1}_m), \qquad \mathcal{L}^k_m = (\mathcal{Z}^k_m)^{\perp}_{\mathcal{C}^k_m},$$

where orthogonality is taken in the graph-space Hilbert norm. Lemma 1 implies that

$$d : \mathcal{L}^{k-1}_m \longrightarrow \mathcal{Z}^k_m$$

is a bounded bijection. Its inverse Q^k is bounded by the bounded inverse theorem. Let $P^k_{\mathcal{Z}}$ be the graph-orthogonal projection onto $\mathcal{Z}^k_m$ and define

$$h^k = Q^k P^k_{\mathcal{Z}}.$$

If $\omega = z + \ell$ is the decomposition in $\mathcal{Z}^k_m \oplus \mathcal{L}^k_m$, then

$$dh^k\omega = z, \qquad h^{k+1}d\omega = \ell.$$

Their sum is ω, proving (2). We set $h^0 = h^{n+1} = 0$ at the endpoints. □

Construction of the Extension

Proof of Theorem 1. The universal extension theorem of Hiptmair, Li and Zou supplies, for every degree, a bounded linear section

$$S^k : \mathcal{X}_m^k(\Omega) \longrightarrow \mathcal{X}_m^k(\mathbb{R}^n)$$

of restriction [3]; this is also the theorem recalled in the introduction of [1]. Restrict its values to K and retain the notation S^k. No cochain property of $S^\bullet$ is used.

For $\omega \in \mathcal{C}_m^k(\Omega)$, define

$$E_m^k \omega := dS^{k-1} h^k \omega + S^k h^{k+1} d\omega. \tag{3}$$

The first term is a closed H^m k-form and hence belongs to $\mathcal{X}_m^k(K)$; the second belongs there by boundedness of S^k. Consequently (3) defines a bounded linear map in the graph norm.

Let r denote restriction from K to Ω. Since $rS^j = I$ and restriction commutes with the distributional exterior derivative, Lemma 2 gives

$$rE_m^k \omega = dh^k \omega + h^{k+1} d\omega = \omega.$$

Moreover,

$$dE_m^k \omega = dS^k h^{k+1} d\omega = dS^k h^{k+1} d\omega + S^{k+1} h^{k+2} d^2 \omega = E_m^{k+1}(d\omega).$$

This proves the first assertion.

Now suppose $\Omega \Subset K$. Choose $\chi \in C_c^\infty(K)$ equal to one on a neighborhood of $\overline{\Omega}$ and replace each whole-space section S^k by χS^k. Multiplication by χ is bounded on every graph space $\mathcal{X}_m^k(\mathbb{R}^n)$, and the new map is still a section of restriction. Formula (3) now has support contained in the fixed compact set $\operatorname{supp} \chi \Subset K$. It therefore vanishes near ∂K, and its extension by zero belongs to $\mathcal{X}_m^k(\mathbb{R}^n)$ with the same bound. This proves the second assertion. □

Verification and Limitations

The proof uses no computation. The identities following (3) are formal; the substantive review point is Lemma 1, specifically the compatibility between the absolute harmonic representatives in [1] and the smooth de Rham cohomology complement in [2]. Both represent ordinary de Rham cohomology on a bounded Lipschitz domain, which makes $P_{\mathfrak{H}}|_{\mathcal{G}}$ an isomorphism.

The result gives bounded operators for each fixed Ω, K, and m. The abstract bounded inverse used to construct h does not by itself give a sharp estimate uniform over a varying class of domains. Thus, if the phrase "analogue of Theorem 1" was intended to demand the same quantitative dependence on a uniform Lipschitz character, that stronger quantitative problem is not resolved here.

Novelty check. On August 9, 2026, bounded searches used the exact source title and close variants of "strict higher-regularity analogue," "higher-regularity cochain extension," $H^{(m,m)}$ with "cochain extension," and "bounded contracting homotopy Sobolev de Rham Lipschitz extension." They found the three papers cited below but no explicit arbitrary-ambient higher-regularity cochain section answering Open question 1, and no occurrence of formula (3) in this setting. Novelty confidence is moderate because the argument is a short synthesis of established analytic and homological ingredients.

Human-review recommendation. Verify Lemma 1 first, then confirm that the intended question does not silently require uniform constants over domain families. Once exactness is accepted, Lemma 2 and formula (3) complete the fixed-domain question.

References

[1] E. Nilsson and S. Pitassi, *Uniformly Bounded Cochain Extensions and Uniform Poincaré Inequalities*, arXiv:2604.04927v2, 2026.

[2] M. Costabel and A. McIntosh, *On Bogovskiĭ and regularized Poincaré integral operators for de Rham complexes on Lipschitz domains*, Math. Z. **265** (2010), 297–320; arXiv:0808.2614v2.

[3] R. Hiptmair, J. Li, and J. Zou, *Universal extension for Sobolev spaces of differential forms and applications*, J. Funct. Anal. **263** (2012), 364–382.